

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) |  |  | <p><b>Indicative content</b></p> <p><b>X</b> identified as hydrochloric acid<br/> <b>Y</b> identified as sodium chloride<br/> <b>Z</b> identified as ethanoic acid</p> <p><b>X</b> is the stronger acid – more vigorous reaction / produces hydrogen more quickly; greater temperature increase / more exothermic</p> <p><b>Y</b> does not react so cannot be an acid / must be sodium chloride<br/> <b>Y</b> must be sodium chloride – magnesium less reactive than sodium so can't displace it</p> <p><b>Z</b> is the weaker acid – less vigorous reaction / produces hydrogen less quickly; smaller temperature rise / less exothermic</p> <p>magnesium + hydrochloric acid → magnesium chloride + hydrogen<br/> <math>\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2</math></p> <p>magnesium + ethanoic acid → magnesium ethanoate + hydrogen<br/> <math>\text{Mg} + \text{CH}_3\text{COOH} \rightarrow (\text{CH}_3\text{COO})_2\text{Mg} + \text{H}_2</math></p> <p><b>5-6 marks</b><br/> All three identified; clear reasoning; good attempt at equation<br/> <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> | 2               |     | 4   | 6     |       | 4    |